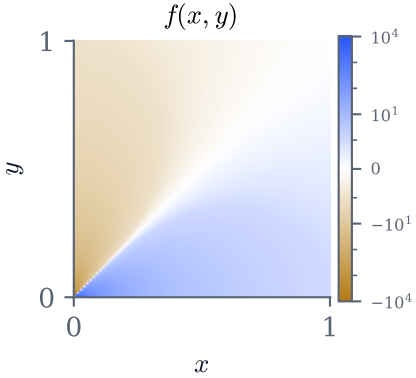
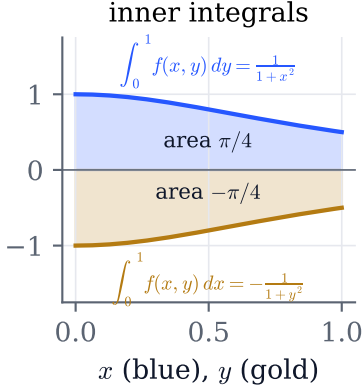
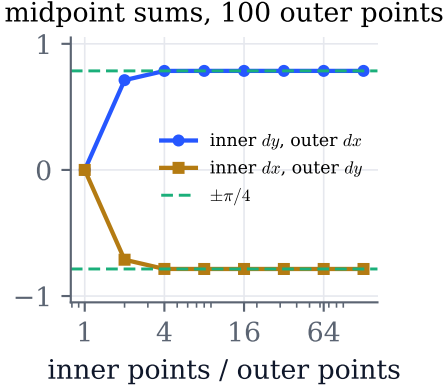

$f(x, y)$
1
0
y
0
1
x
10^4
10^1
0
-10^1
-10^4


inner integrals
$\int_0^1 f(x, y)\, dy = \frac{1}{1+x^2}$
area $\pi/4$
area $-\pi/4$
$\int_0^1 f(x, y)\, dx = -\frac{1}{1+y^2}$
1
0
−1
0.0
0.5
1.0
x (blue), y (gold)


midpoint sums, 100 outer points
inner dy, outer dx
inner dx, outer dy
$\pm\pi/4$
1
0
−1
1
4
16
64
inner points / outer points